Chassis User Manual

Standard version of the universal chassis (as shown in the picture below)

Table of contents

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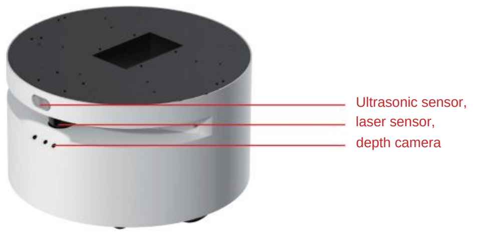
Performance Parameter Table

Indoor general-purpose robot chassis parameters		
category	**parameter**	**index**
Basic parameters	size	D500*H310mm
	weight	35kg (10kg counterweight optional)
	No-load battery life	12h
	Charging time	4h
	Maximum load	50kg
	Battery capacity	24V/20AH
	Chassis type	2* Differential Drive + 4* Omnidirectional Wheels + 1* Fixed Wheel
	Main hardware configuration	1 x 20m LiDAR TOF
		6.5-inch servo hub motors (with odometer) *2
		RGBD Depth Camera (0.6-8m) *1 (Optional, not included in standard version) Ultrasonic
		Ranging Sensor (0.3-1.2m) *1
		IMU six-axis attitude sensor *1
		High-performance X86 industrial computer *1
Maneuver parameters	Maximum navigation speed	0.8m/s
	Maximum speed	1.5m/s
	Climbing angle	5°
	Obstacle crossing height	10mm
	Ditch width	40mm
	Turning radius	0mm
	Maximum rotational speed	60°/s
	Positioning accuracy	±5cm
	Mapping area	10000m2
	Navigation accuracy	±5cm
Extended Interface	Power interface	24V (200W)
	USB	Two, with reserved peripherals.
	Network interface	RJ45, for communication or routing
	Physical installation interface	Structural mounting holes
	Other interfaces	System switch, emergency stop switch
Charging pile parameters	Charging method	Contact charging
	Input voltage	AC 110-240V
	Output voltage	29.4V
	Output current	7A
	Protection mechanism	Short circuit, overcurrent, etc.
Environmental requirements	Work environment	Temperature: 5ÿ-40ÿ
	Storage environment	Temperature: -10ÿ to 60ÿ

Basic usage

1. Power on/off operation



1.1 Main power switch

The machine has a main power switch located at the bottom, next to the right-hand caster wheel on the rear. This switch is used to disconnect the battery power. It is off by default at the factory and needs to be manually switched to the on position upon first use. If the machine will not be used for an extended period

(more than one month), the battery level should be maintained at around 40%, and the main power switch should be switched to the off position to prevent the battery from becoming depleted due to prolonged inactivity. (See picture)

1.2 Powering on

Touching the power button once will trigger a short buzzer sound, followed by five consecutive buzzer sounds after a short period of time, indicating that the chassis is starting up and power is supplied to the external power interface.

(As shown in the picture)

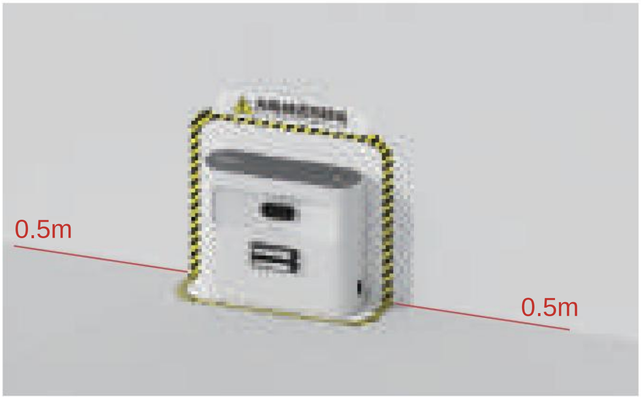
1.3 Power off

Press and hold the power button until the buzzer sounds continuously, then release and wait for the machine to turn off.

2. Machine charging

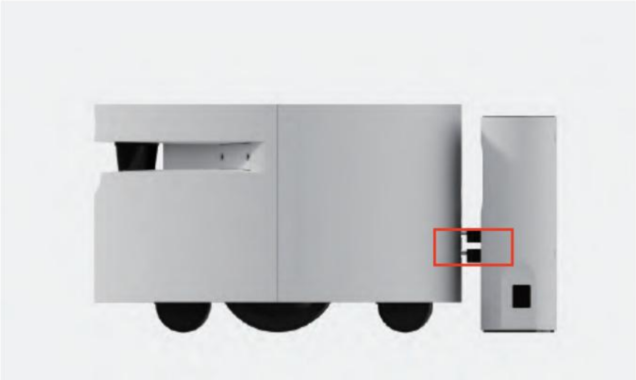
2.1 Placement of charging stations

(1) The charging station needs to be placed against a wall, and there should be no other objects within half a meter on either side of the center of the charging station. (See figure)



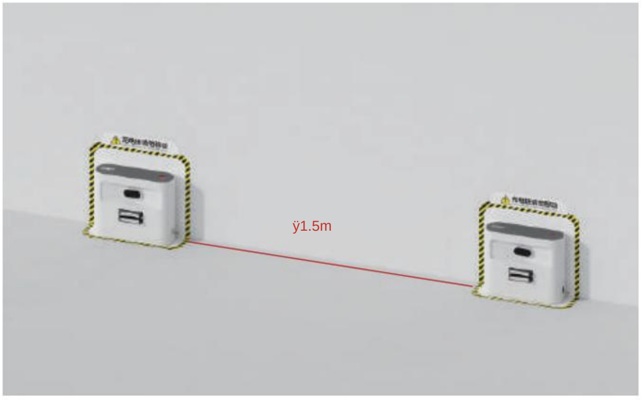
(2) The charging station needs to be placed on a flat ground and cannot be placed on a slope or in a carpeted environment.

(3) The height of the charging pad on the charging pile needs to be the same as the height of the charging contacts on the machine. If there is any error, the bottom of the charging pile needs to be adjusted, and the filling material needs to be increased or decreased appropriately. (See figure)



(4) The location of the charging station needs to be relatively fixed to the surrounding environment to avoid movement due to personnel or other reasons later. It can be secured by affixing markings to the ground or using double-sided tape. (See figure)

(5) When multiple machines are used simultaneously in the same scenario, the distance between charging stations must not be less than 1.5m. (See figure)



(6) After ensuring the charging station is properly placed, connect the charging station to the power source using the power cord. At this time, a red indicator light will illuminate on the charging station. (See figure)

2.2 Contact charging

(1) After the charging station is fixed in position, manually push the machine to the charging station and make the charging contacts on the bottom of the machine contact the charging pads on the charging station to start charging normally. After normal contact, the charging station indicator light will turn green. (See figure) (2) When the map is normal and the charging station location has been marked, you can send a charging station location task to make the machine automatically return to the charging station location and automatically identify and recharge. (3) When the map is normal and the charging station location has been marked, you can send a charging station location task to make the machine automatically return to the charging station location and automatically identify and recharge. (4) If you need to charge while the machine is off, you need to turn on the main power switch.

Note: The charging station indicator light has only two states: Red: The charging station is powered on but not in contact with the device for charging. Green: The charging station is powered on and in contact with the device for charging, and the machine's battery level is less than 90%. The charging station light will turn green when the machine is charging, and this also applies when the machine is off and charging. If the light does not turn green when the machine is on the charging station, please check if the charging station is powered on properly and if the machine is making proper contact with the charging station.

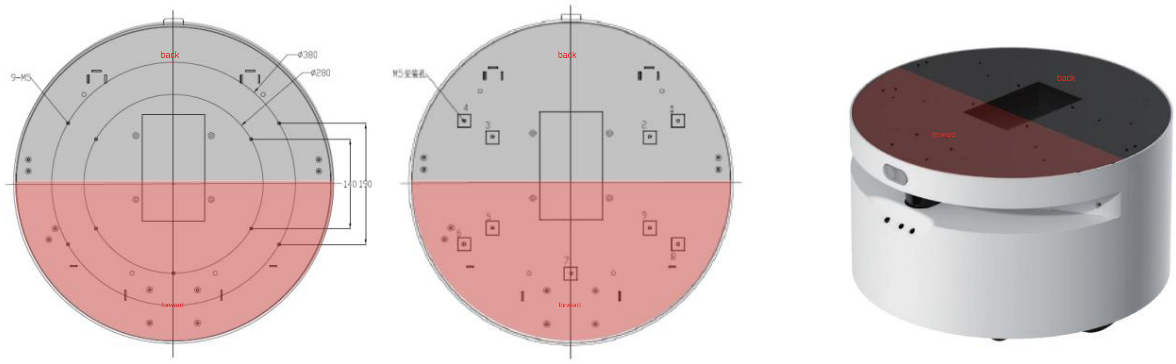
User Guidelines 1.

Overall Design Recommendations

(1) Except for the chassis, the upper body height is recommended to be below 90cm. If it is higher, the chassis cannot guarantee the balance of the whole machine when crossing obstacles and climbing slopes. At the recommended height, the equipment's ability to climb slopes and cross obstacles can reach about 85% of the standard value. If the height exceeds the range, it cannot be guaranteed. (2) The center of gravity of the whole machine should not exceed 50cm above the ground. If the center of gravity is higher than 50cm, the whole machine will shake violently or tip over when crossing obstacles and climbing slopes. (3) The diameter of the whole machine is recommended to be kept within the range of 50cm + 3cm of the chassis diameter. If it is larger than this range, you need to contact our technical staff to adjust the software to adapt it. Various parameters of the equipment, such as climbing, crossing obstacles, minimum passing distance, etc., need to be recalculated. (4) If you need to make changes to the chassis structure, you should keep the sensor position unchanged as much as possible, and the shell should not block or interfere with the sensor. If you need to change the sensor position, please contact our technical support personnel first to communicate and confirm before determining the modification plan. (5) It is recommended to add an overhead camera to the whole machine to avoid three-dimensional obstacles in the blind spot of the chassis laser, and at the same time avoid the risk of falling.

1.1 Upper body structure assembly

The chassis has nine pre-drilled M5 mounting holes, which are symmetrically arranged. (See figure)



Use screws approximately M5 x 10mm in diameter; do not use screws that are too long to prevent them from encroaching into the internal space of the chassis and causing chassis malfunctions or damage.

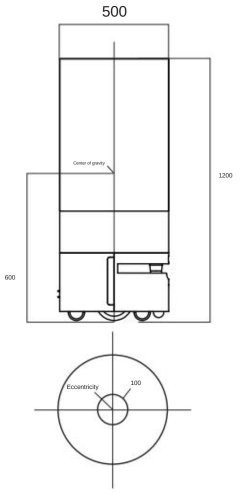
1.2 Upper Body Design Dimension

The upper body design must not obstruct the sensors around the chassis. If it does, you need to confirm and communicate with the technical staff.

1.3 Upper Body Center of Gravity Height (1)

Excluding the chassis, the recommended overall height of the upper body is ÿ90cm. An excessively high upper body cannot guarantee the safety of the entire machine and its ability to overcome obstacles, climb slopes, and pass through gaps. In severe cases, it may cause the machine to tip over while climbing slopes or sway back

and forth. (2) Recommended upper body design example, as shown in the figure below, with an upper body center of gravity height of 550mm. The design center of gravity height should be less than the center of gravity shown in the figure. It is recommended that the design center of gravity height be less than or equal to 500mm, the axial eccentricity range be <ÿ100mm, and the upper body weight be designed for 50kg. This can achieve 85% of the ability to overcome obstacles, climb slopes, and pass through gaps (as shown in the figure).



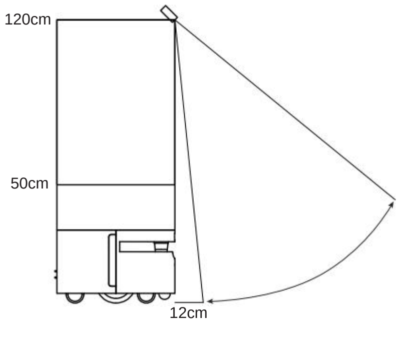
1.4 Modification of upper-body camera

(1) The current equipment is suitable for obstacle avoidance using laser and chassis-mounted upward cameras. It cannot detect obstacles below laser level or in fall scenarios. Therefore, a downward-looking camera needs to be installed on the upper body of the robot to avoid low obstacles and fall scenarios. (2)

Installation height of the upper body camera: The recommended height is 50~120cm. If the height is lower or higher than this, the equipment will not be able to properly identify obstacles in front of it. The recommended vertical angle is 32°. The specific settings depend on the camera's parameters and performance. (3)

Installation angle of the upper body camera: First, determine the camera's viewing angle. Using the front of the robot as a reference, install cameras at the front and back, and at the center of symmetry in the left and right directions. Second, adjust and calculate the camera's shooting angle so that it can see the closest distance in front of the robot's chassis (as shown in the figure), and there should be no obstruction from its own structure. (4) On the

depth camera, the side with the connecting cable inserted is the bottom. If it is installed backwards, it will cause the image to flip.

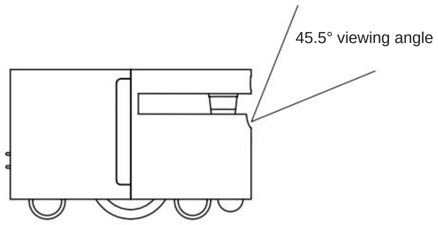


After the camera is designed and installed, it will only take effect after our technicians adjust the parameters. Each device needs to be configured individually.

1.5 Depth Camera

(1) The depth camera has a 45° field of view. The original camera assembly maintains a 51° angle with the horizontal plane. For any changes in requirements, please contact our technical staff. (See figure) (2)

This depth camera is suitable for obstacle avoidance using lasers and chassis-mounted upward-facing cameras. It cannot properly detect obstacles higher than lasers or horizontal bars (such as tables). Therefore, an upward-facing camera needs to be added to the upper part of the device to avoid high obstacles in non-wall scenarios. (Only depth cameras provided by our company can be used.)



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2. Environmental survey before use

The mobile chassis product is only used in flat indoor environments with no height difference. In view of the blind spots of sensor detection and the interference of different object materials on the sensor, the following lists the characteristics of the corresponding stable operation scenarios and information on dangerous scenarios. Before the equipment is brought to the site, it is necessary to conduct a corresponding survey to determine whether it is suitable for the operation of the equipment. (1) The equipment can only be operated in an indoor environment. (2) The equipment operation site must

ensure that there is no height difference and that the height difference

between the same plane in the environment does not exceed 1.8cm. (3) If there is a slope in the operation site, it must be within the range of the equipment specifications and the height of the whole product does not

exceed 110cm. (4) When there is a lot of glass in the environment, such as glass walls, glass doors, etc., the shielding stickers provided by our company should be affixed to the height of the lidar in the glass

environment. (5) If there is a risk of falling, slope, normally closed fire doors, etc., the upper body downward view camera must be installed in the environment where it is easy to fall, and it must be installed within the specified angle to avoid the equipment not being recognized and causing a fall.

(6) If there are frequent renovations or environmental changes in the operating site, the map should be rebuilt in a timely manner to avoid the risk of map offset during equipment operation. (Risk areas include, but are not limited to: fire escape stairs, level stairs, small steps, and escalators).

2.1 Fall Risk

(1) When a site has a drop of more than 1.5 cm below ground level or is a non-planar environment, there is a risk of equipment falling or tipping over during movement. (2) When deploying in a site with a risk of falling, an environmental safety assessment should be conducted in accordance with the "Self-developed Chassis Deployment Survey Specification," and the results should be statistically reported through the "Foundation Leveling Deployment Survey Document," as follows:

The self-developed chassis deployment survey

specification document for foundation leveling deployment survey is as follows:

机器人服务环境评估表(2021/6)				
A、勘察项（所有勘察项都要按要求留存现场的照片）			答案	备注
1	整体情况	完整项目名称		
2		详细地址（含完整门牌号）		
3		对接人（客户）		
4		联系方式（客户）		
5		现场计划部署几台机器人？		
6	网络通信	移动、电信、联通信号哪个最强？哪个最弱？（请按顺序排名）		
7		是否可提供稳定可连外网的WIFI网络		
8	项目场景	A、写字楼 B、酒店 C、医院 D、政务 E、展馆 F、商超 G、其他（自填）		
		A、室内 B、室外（含露天区域）		
9	部署面积	场景面积(项目场景面积)	m^2	附视频
		部署面积(机器人实际行走路径所占面积)	m^2	自研底盘活动面积不可大于1000m^2（附视频）
		现场环境有无装修范围（大面积围挡）？		
10	地面类型	A、瓷砖 B、大理石 C、水泥 D、木板 E、地毯 D、其他（自填）		
11	地毯类型（否略填写）	材质:A、纯毛 B、纤维混纺 C、化纤 D、草编 E、其他（自填）		
		厚度:A、1cm以下 B、1cm-2cm C、2cm以上		
12	地面勘察	是否存在减速带、地标灯？多少处、长多少、宽多少（模板：2处/3cm/2cm）		
13		行走路径是否有临时障碍物？（桌椅、花盆、围栏等，填写物体及数量）		
14		是否存在向下楼梯、台阶		
15		是否存在上下行自动扶梯		
16		是否存在高度差>15mm的台阶		轮式机器人，通过性能较大依赖于地面平整度
17		是否存在角度>5°的斜坡		轮式机器人，通过性能较大依赖于地面平整度
18		机器人行走路径中是否存在大面积玻璃区域		
19		机器人活动区域是否存在有跌落风险的常开/常闭安全门		常开安全门、常闭安全门后有跌落风险区域的都不可部署
20		是否存在高度差>1cm的凹坑、坎		，通过性能较大依赖于地面平整度
21	走廊	最大宽度和最小宽度（模板：10m/1m）		
22	玻璃区域	是否有大面积玻璃区？有多少处？长度多少？是否可以做处理？（模板：是/2处/5m/不可以）		

A1、多机运作确认事项			答案	备注
23	多机运作	是否一个区域存在多台机器或多款机器？请列举说明（欢欢云迹底盘 x1/消毒自研底盘 x2/欢欢云迹 x1、消毒自研 x1）		
		是否工作在同一个区域？		
		区域面积多大？		
		是否工作在同一个时间段内？		
		工作点是否发生重叠或者临近？		
		行走路径是否发生交叉或者临近？		
		能否通过禁行线方式分割多台机器人工作区域？		

2.1.1 Overall Situation

(1) The complete project name, detailed address, contact person, and contact person can be filled in according to the contents of the work order. The accuracy of the information should be confirmed with the client on site to avoid problems during the connection. (2) The

number of robots to be deployed should be determined and written down after the on-site survey.

2.1.2 Network Communication

(1) On-site check for dead zones in the mobile network signal. If the mobile robot is in an area with no signal, it may go offline. (2) For the on-site WiFi, determine whether an account

connection or verification code is required. If so, the robot may be unable to verify and go offline.

2.1.3 Project Scenario

(1) Record the scene in which the machine is deployed according to the on-site environment.

(2) Record whether it is used outdoors or indoors.

2.1.4 Deployment Area

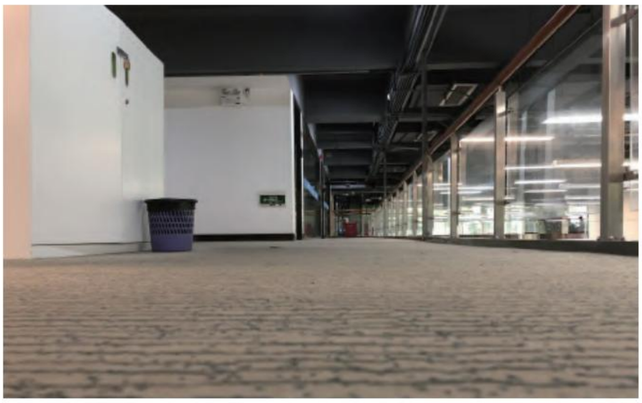
(1) Scene area: The floor area where the project is located. (2)

Deployment area: The operating area of the machine. (3)

The site environment needs to be assessed to determine if there are any fenced-off or renovation areas, as changes to the scene will require redrawing the plans. Therefore, if there are fenced-off areas, the client needs to be informed of the precautions in advance. (See figure)

2.1.5 Ground Type

(1) Fill in the information according to the type of ground surface at the site. (See figure)



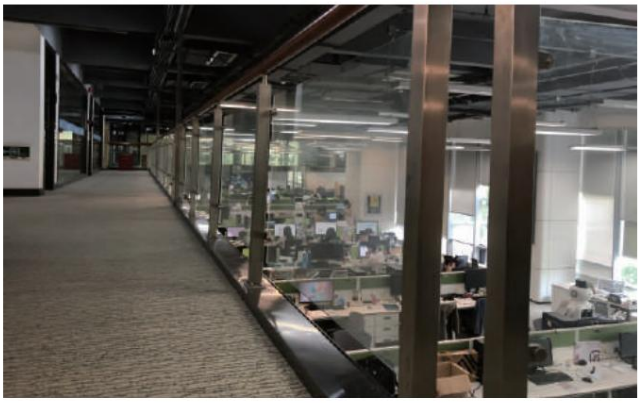
2.1.6 Ground Survey

(1) Check if there are landmark lights on the ground. If the landmark lights are too high, it may cause the machine to shake and stop spraying. (See picture)

(2) Check if there are too many temporary obstacles on the ground, as too many obstacles can cause the machine to shift position. (See figure)

(3) Check if there are any downward or upward steps or stairs in the environment. This is quite important; if there are downward stairs or steps, the machine may tip over, and deployment is generally not recommended. (See figure)

(4) Check if there are continuous large glass surfaces in the operating environment. If so, a protective film needs to be applied; otherwise, the safety of the machine's operation cannot be guaranteed. (See figure)



(5) Check the environment for steps with a height difference of 15mm or more, or slopes greater than 5°. Machine operation is risky in such environments, and deployment is generally not recommended. (See figure)

(6) Check if there are any normally open or normally closed safety doors in the environment. Safety doors usually contain stairs leading up or down, which also constitutes a fall risk area. (See figure)

(7) Check the environment for pits and bumps with a height difference of 1 cm or more. (See figure)

2.1.7 Carpet Types

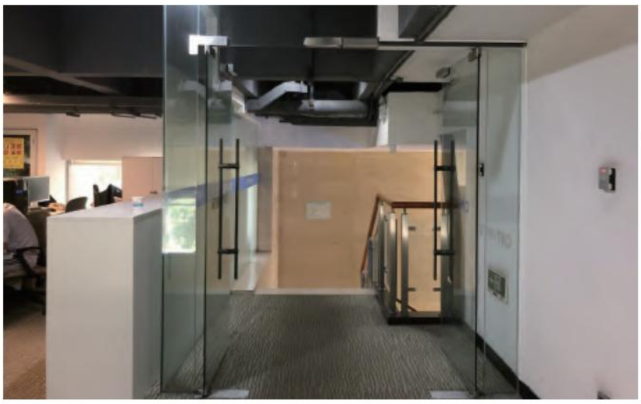
If there is carpet, its material and thickness need to be determined. Excessive thickness will create a height difference between the robot's charging contacts and the charging dock, causing misalignment or preventing it from returning to the charging dock. (See figure)

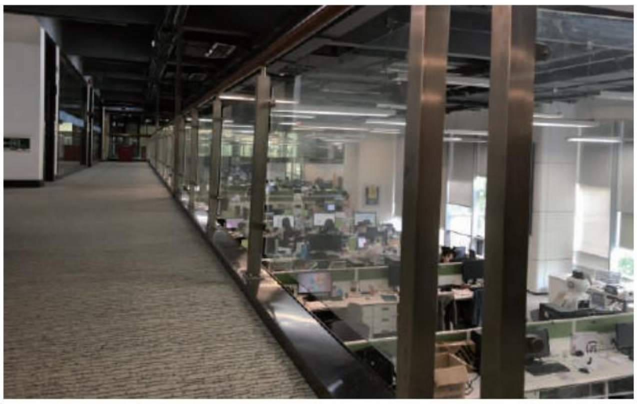
2.1.8 Corridor

Check the minimum width of the corridor in all directions. (See diagram)

2.1.9 Glass Area

(1) Check if there are any continuous large glass surfaces or glass doors in the operating environment. If so, a protective film needs to be applied; otherwise, the safety of the machine's operation cannot be guaranteed. (See figure)





2.1.10 Drop environments include, but are not limited to, those mentioned above.

When the environment contains a potential for falls, it is not recommended to move the equipment on-site if the equipment is not equipped with a top-view camera. If a top-view camera is installed, an environmental survey must be conducted in accordance with section 2 and reported. (1) Stairs (downward), escalators (upward, downward). (2) Steps with a drop of more than 1.5cm.

(3) Environments with steep slopes (above 10 degrees, without frictional ground slopes) and steep ledges (above 1.8 cm).

(4) Fire safety passage (normally open door, normally closed door): There are environments such as fall-off areas and fire stairs behind the door.

2.2 Map offset risk

Environmental

reasons: (1) When there is a carpet with a thickness of more than 0.5cm in the on-site operating environment, there is a high probability of causing the equipment map to shift.

(2) When used in icy, frosty, or slippery environments, there is a high probability of map shift.

(3) When moving in a large open area beyond the laser recognition range without reference objects, there is a high probability that the equipment map will be offset.

(4) Significant changes in the environment of the operating site compared to the last map: such as renovation, if the layout is changed and no new map is created, there is a high probability that the equipment map will be offset.

(5) Operation in complex environments: If there are many moving tables and chairs in the office and they are frequently changed, there is a high probability that the equipment map will be offset.

(6) When the site environment has slopes, ridges, ditches, etc., and the speed is relatively fast when passing through, there is a high probability that the equipment map will be deviated.

(7) When moving, the sensor cannot identify obstacles: such as obstacles that are more than 20cm above the ground when the top-view camera is not installed. This can cause bumps and scrapes, which may lead to map shifts in the device. (8) The device needs to be turned on at the charging station or touched before it can be used.

Reason for operation:

(1) When using tools or interfaces incorrectly for position correction, there is a high probability of causing the device map to shift.

(2) After the device is turned off, push it to another position and turn it on without charging.

(3) The device is charging/powering on at a charging station, but the location of the charging station does not match the actual location.

(4) The wheels lift off the ground and move when the equipment is powered on.

(5) Pushing or dragging the equipment for an extended period of time when it is powered on, except in an emergency stop.

2.3 Collision Risk

(1) No overhead camera installed: Obstacles less than 23cm below the ground. Overhead camera installed: Obstacles less than 8cm below the ground. No overhead camera installed...

(2) Objects made of reflective, light-absorbing, or transparent materials.

(3) Visible objects with a volume of less than 2cm or a protrusion of less than 2cm.

(4) If the overall height of the equipment is higher than 1.1m and no additional configuration is made, there is a probability of colliding with objects that are higher than 1.1m but lower than the height of the equipment.

2.4 It needs to be marked on the map.

(1) In areas where the equipment is not allowed to enter, virtual walls must be marked to prevent unauthorized access.

(2) If the mapping area is larger than the actual operating range of the equipment, make sure that the virtual wall surrounds the actual operating range as much as possible to avoid excessive movement of the equipment due to map offset.

(3) If there is a dangerous area in the area, the virtual wall should be at least 0.5m away from the dangerous area. (4) If the object actually exists in the scene and can be detected by radar, but is missed in the actual scan, it needs to be filled in with a no-entry zone. (5) In glass environments, such as glass walls and glass doors, the area needs to be marked with a virtual wall.

(6) At three-dimensional obstacles above the radar scanning height, the area needs to be marked with a virtual wall.

(7) At low obstacles, such as bumps, pits and carpeted areas that do not meet the chassis performance parameters, the area needs to be marked with a virtual wall.

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Tool usage

1. Establish communication

1.1 Wireless Communication - WiFi

The device has a built-in heating element. To connect to the WiFi hotspot, check the WiFi list and select the one named TY1251D-xxxxx (where the last five digits represent the last five digits of the device's serial number). The default password is 123456789. Ensure that the device's wireless network card is not configured with a static IP address. Once connected to the WiFi, the connection is successful. (Multiple devices can connect to the same device.)

1.2 Wired Communication - Network Cable

The robot gateway is 192.168.20.1, and the IP address is 192.168.20.22. There is no automatic IP assignment; the peripheral device needs to configure it manually.

2. Mapping Tools 2.1

Chassis Connection

Step 1:

Switch the control terminal to chassis WIFI. Details are as follows:

SSID: TY1251C003-xxxx (This WiFi name is related to the chassis hardware version and the project)

Initial password: 123456789

The default IP address is: 192.168.20.22

Default IP: 10.42.0.1

For tablets like Huanhuan, which are Android tablets with built-in disinfection features, a wired connection can be used.

Step Two:

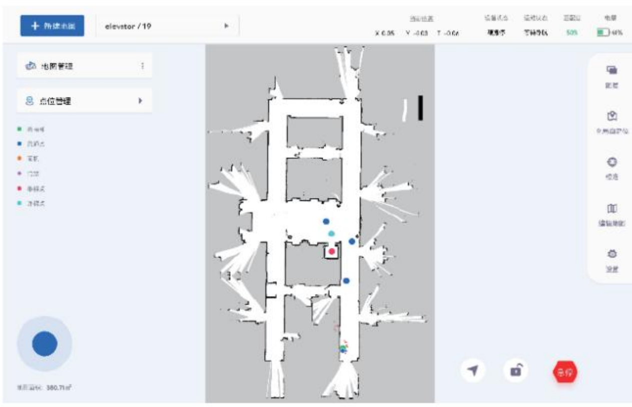
Click the icon to open the self-developed chassis deployment software.

Step 3:

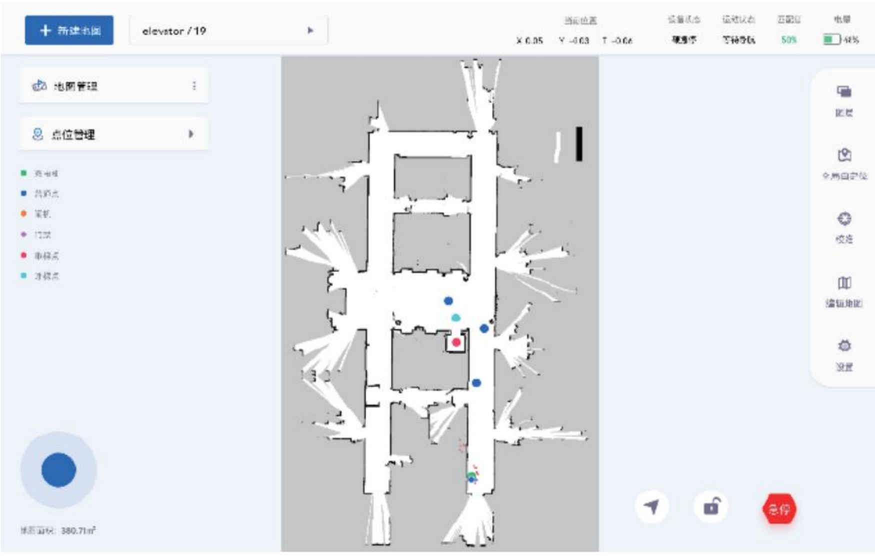
Enter the chassis's IP address (default IP: 10.42.0.1). If it's an Android tablet connecting directly, use 192.168.20.22. After entering the IP address, click "Connect Chassis".

Step Four:

The following image will be displayed after a successful connection.



2.2 Interface Description (as shown in the figure)



2.2.1 The top displays machine status information, allows you to create a new map, and lets you select a map.

Robot status information includes current position, device status, and motion status.

And compatibility and battery level.

Click the button  on the left to access the new map creation function.

Click on the map name to enter the list and switch the map graphics displayed on the main map.

2.2.2 The left side displays map management and location management.

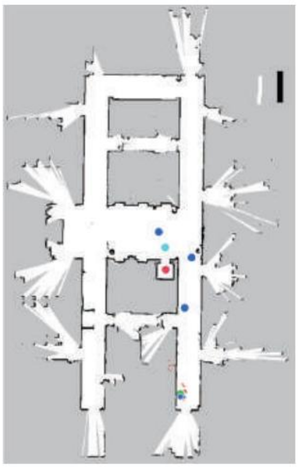
Click on [Map Management] to expand the map list. Click on [Location Management] to expand the location list.

2.2.3 The right side shows map-related functions.

This section includes map layers, global auto-localization, calibration, and map editing.

And settings functions.

2.2.4 The middle area displays the information of the currently selected map graphic.



The current map interface supports two-finger operation to zoom in, zoom out, and drag the map.

2.2.5 The bottom section contains robot control-related functions.

On the left is the manual remote control button for the robot, which controls the robot's direction of movement.

The current map area is displayed at the bottom left. The buttons on the right are for setting points, respectively.

Navigation function, lock/unlock remote control function button, software emergency stop button

2.3 Creating a New Map

Step 1:

Click the button to enter the new map page. Fill in the [Map Name], [Map Floor], and select the [Map Mode].

Standard Scene: Standard scenes are generally used for mapping.

Similar Scenes: If the mapping environment consists mostly of similar corridors, narrow passages, etc., with little variation in reference objects, it is recommended to use the "Similar Scenes" mode

for mapping. Large Scenes: If the mapping environment is around 1000-3000 square meters, it is recommended to use the "Large Scenes" mode for mapping.

Step Two:

After confirming the newly added map information, enter map scanning mode.

Blue arrow: Indicates the robot's location and orientation. Red dot: Represents radar point cloud data.

White area: Obstacle-free area. Gray area: Unknown

area. Black area: Obstacles. Complete

scan (click to cancel scan).

Click the button to save the map successfully. Click the button again to prevent the map from being saved.

Use the joystick to control the chassis direction and perform map scanning. Alternatively, you can use manual pushing (recommended). After the robot presses the hard stop button, manually push the robot to perform mapping.

• Mapping instructions:

1. Use manual pushing as much as possible, and keep the speed slow and gentle. Avoid excessive rotation speed. 2.

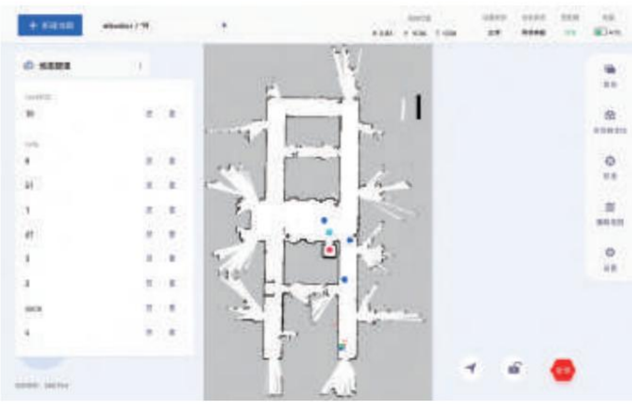
The starting location for mapping doesn't necessarily have to be a charging station, but the starting location should have an open environment

with clearly visible features. 3. Complete the large loop sections first, and then add details last.

2.4 Map Management Module

Step 1:

Click on [Map Management] to expand the map list, which supports editing and deleting maps.



Step Two:

Click the button to enter the map editing page. You can modify the [Map Name] and [Map Floor] information.

Machine Translated by Google

2.5 Location Management Module

Click on [Site Management] to expand the site list, which supports adding sites, site navigation, site editing, site deletion, and cross-floor navigation.

2.5.1 Adding new locations

Click on "Add Point", enter the information for the added point as shown in the image, and click "OK" after entering the information. The point has now been added.

• Location Notes: 1.

Currently, only one charging station can be set up per location to prevent conflicts. 2. The turnstiles and

access

control systems at the locations require corresponding hardware modules (not included by default). 3. The

elevators

at the locations require corresponding hardware modules (not included by default).

2.5.2 Point Navigation

Select the desired location from the list of locations and click the button to navigate to that location.

2.5.3 Cross-floor navigation

Click on the "Cross-Floor Navigation" pop-up window to display a list of locations on other floors. Select the

corresponding location and confirm to execute the navigation operation to perform the cross-floor navigation

task.

1. Before executing cross-floor navigation tasks, elevator control information needs to be configured, and the

algorithm version must be 4.5.1 or higher. 2. The

boarding point needs to be positioned facing the elevator door and located in the center inside the elevator;

the calling point needs to be facing the elevator door and located directly in front of the elevator.

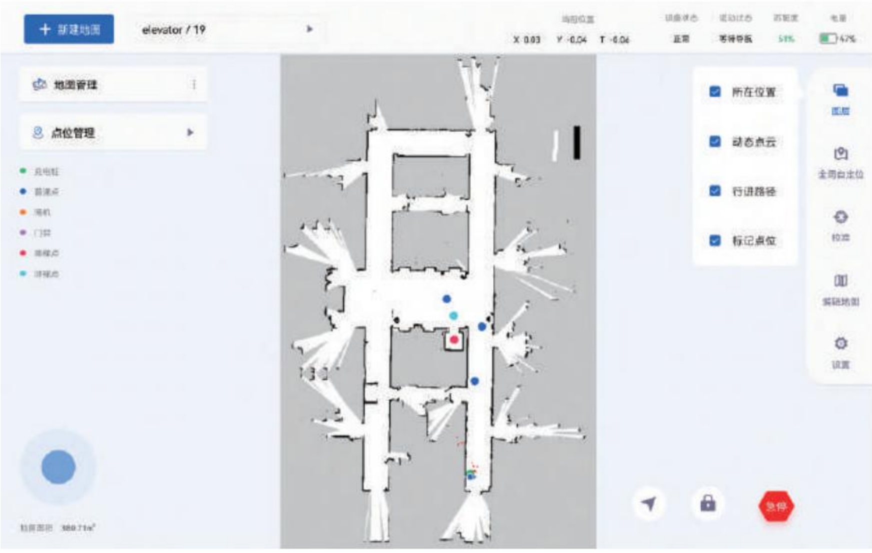
Note: Tap the navigation status prompt box at the bottom center of the screen 6 times consecutively to

forcibly cancel elevator control.

Machine Translated by Google

2.6 Layer Management Module

Click the [Layers] button to expand all layers of the current map. The selected layer will then display its information on the map.



2.7 Global Self-Location Module

Click the "Global Auto-Localization" button to bring up the information menu, then click "OK" to automatically correct the robot's position on the map.

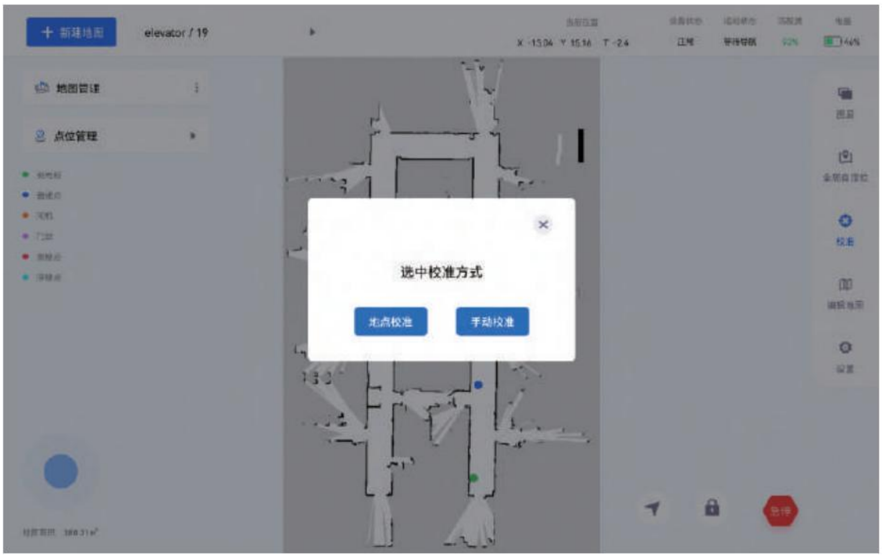
• Function

Description: 1. When activating this function, personnel should stand behind the robot. Standing in front and obstructing the robot's radar scanning area can easily lead to positioning errors. 2. When activating this function, the robot scanning environment should be as open and feature-rich as possible. 3. This function has a certain success rate and is time-consuming. It will not always be correctly positioned. The accuracy depends on the environmental features and map size. 4. If the positioning function fails, the manual positioning function can be used to set the robot's position.

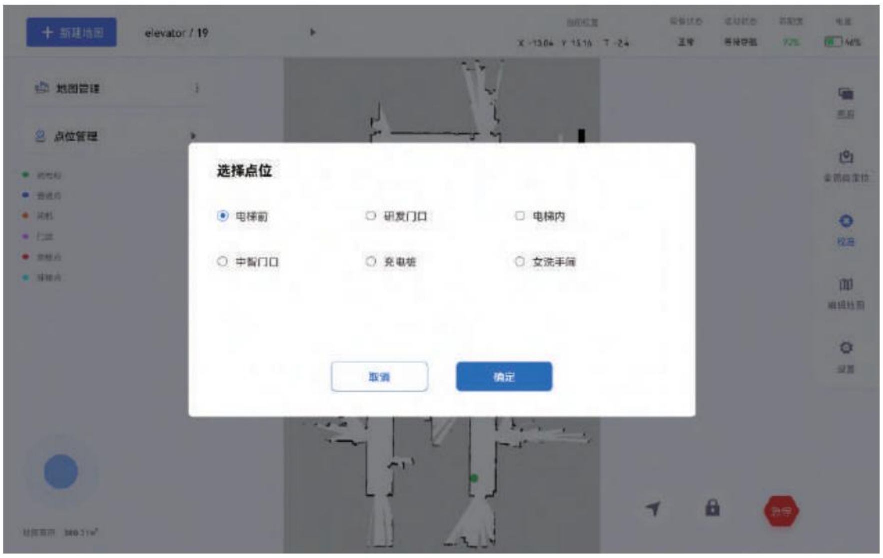
Machine Translated by Google

2.8 Calibration Module

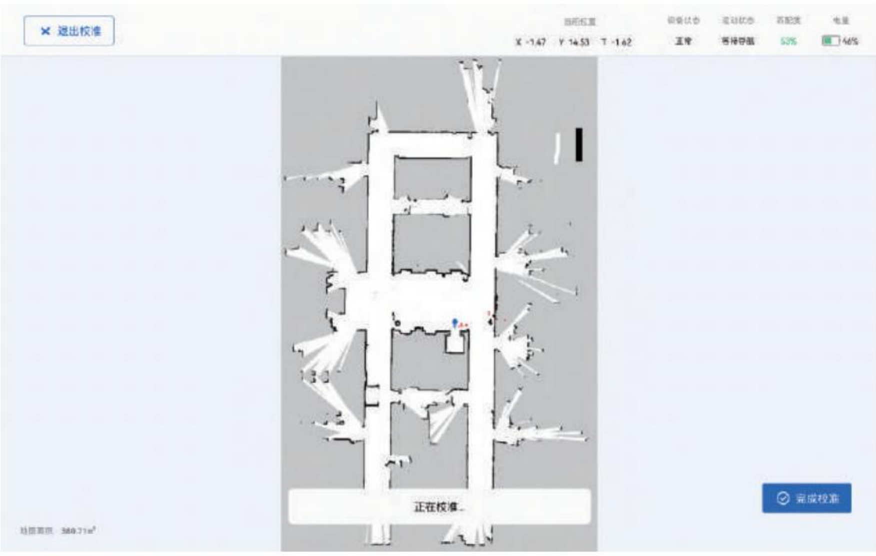
(I) Click the [Calibration] button, and a pop-up window will appear as shown in the figure. You can choose between two calibration methods: [Location Calibration] and [Manual Calibration].



(ii) Selecting ÿLocation Calibrationÿ will take you to the point list. Select the point and click ÿOKÿ to start the location calibration.

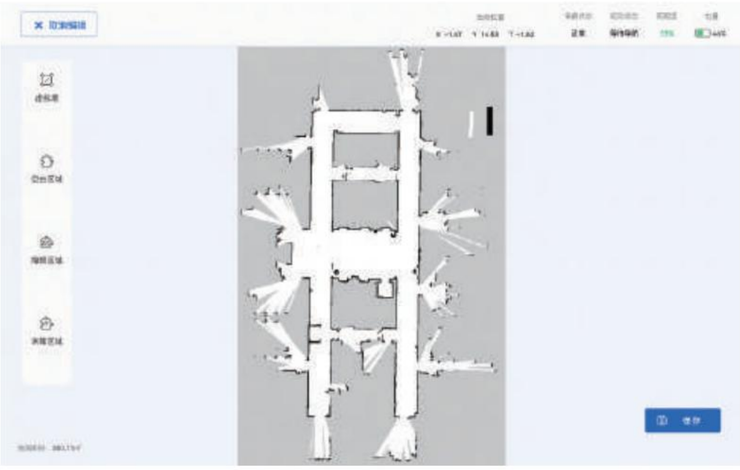


(III) Select ÿManual Calibrationÿ to jump to the map page, select a point on the map, specify the direction, and start calibration.



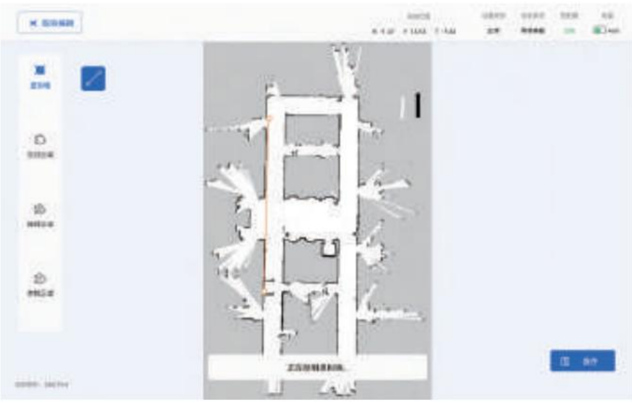
2.9 Edit Map

Click the "Edit Map" button to enter the map editing interface. Once you've finished drawing, click "Save".

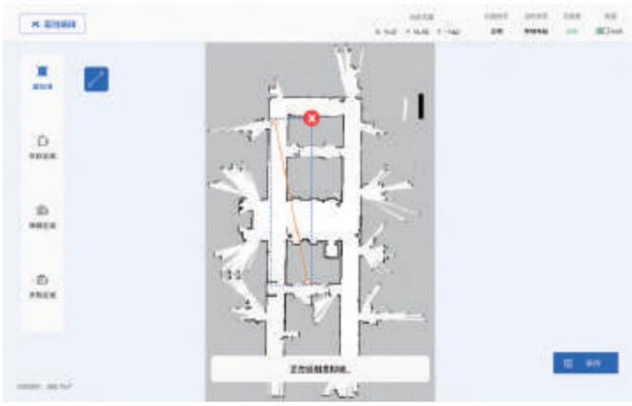


2.9.1 Virtual Wall Drawing

Click on [Virtual Wall] to enter the virtual wall drawing page, and select "Start Drawing Line Segments". 

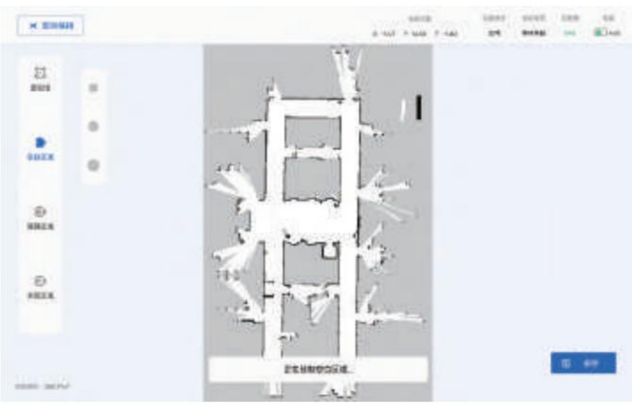


Selecting a drawn virtual wall on the map will bring up an editable operation box. As shown in the image, dragging the dots at both ends of the line segment allows you to modify the length and position of the line segment. Clicking will delete the virtual wall.

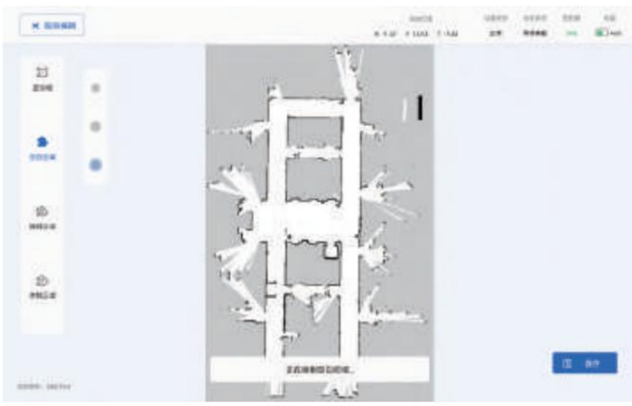


2.9.2 Region Drawing

Click on the [Blank Area], [Obstacle Area], or [Unknown Area] to enter the drawing page for that area.



After selecting the brush size, you can draw the corresponding area.



2.10 Configuration Module

The settings module includes three modules: chassis configuration, version information, and fault diagnosis.

2.10.1 Chassis Configuration

(a) Operation settings:

Operation settings include speed configuration, travel mode, and manual remote control configuration.

The speed settings only affect the robot's autonomous navigation speed and do not affect manual operation.

Speed Level - High: Faster movement speed;

Speed Level - Medium: Moderate movement speed (default robot speed); Speed Level - Low: Slower movement speed; Movement Mode - Safe: Larger obstacle avoidance distance, cannot pass through narrow paths;

Movement Mode - Balanced: Moderate obstacle avoidance distance; Movement Mode - Efficient: Smaller obstacle avoidance distance, can pass through narrow paths; Direction Mode - Four-way arrow keys: Traditional directional keys, as shown in the image; Direction Mode - Eight-way arrow keys: Traditional joystick, as shown in the image.

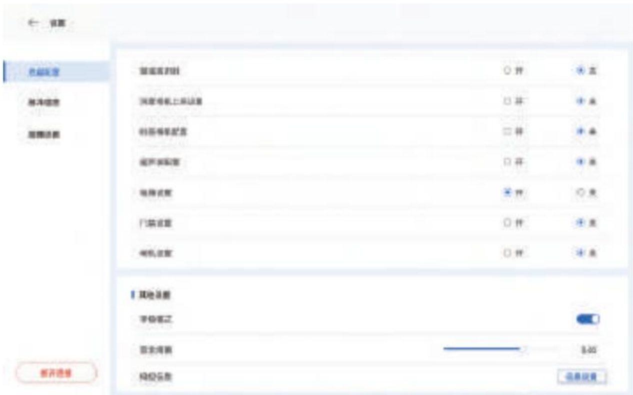
(II) Sensor Configuration:

The sensor configuration module supports configuring the switches for the sensors shown in the image below. After making your selections, click the "Save Settings" button. A pop-up window will prompt you to return to the charging station and restart the chassis, as shown in the image. Push the robot to the charging station and restart it for the settings to take effect.

Note: Before configuring this module, please check whether the robot itself is equipped with the relevant sensors.

(III) Other settings:

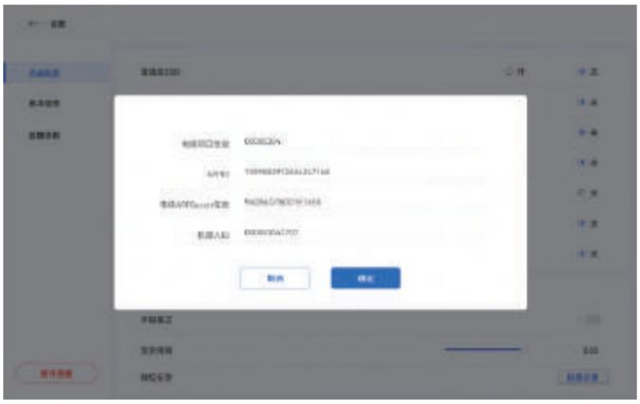
Other settings include smooth mode, safety thresholds, and elevator control information configuration. Click the "Information Settings" button for elevator control information to enter the settings page.



Stable Mode: Enabling stable mode enhances the robot's stability during movement and docking. You can choose whether to enable it based on the usage scenario. Safety Threshold: During movement tasks, if the matching degree falls below the safety threshold, the robot will spin in place and automatically stop after 10 seconds. Please set the safety threshold appropriately based on the environmental matching degree to prevent the robot from continuously spinning in place.

Note: Speed configuration, travel mode, elevator configuration, sensor configuration and other functions require algorithm version V4.5.0 or above to function properly.

As shown in the figure, elevator control information needs to be configured in order to perform cross-floor navigation tasks.



Version information 2.10.2

The algorithm version can be updated via USB drive or online. It also supports chassis driver firmware updates and industrial PC time updates. After the chassis algorithm update is complete, please power off and restart the robot.

After the underlying firmware version update is complete, the robot will automatically power off. Do not perform any operations before the robot automatically powers off. After the robot automatically powers off, you can power it on normally.

After updating the industrial PC time, please power off and restart the robot.

2.10.3 Fault Diagnosis

Click "Start Detection" to begin the self-detection process and enter the self-detection results page to view the results.

2.10.4 Disconnect

2.11 Operating Area

: Controls the chassis's start, stop, and direction of

travel. : Click the icon to enter the point navigation page. Click the target point on the map and specify the direction; the robot will then move to that point. : Performs an

emergency stop on the chassis via software. The robot cannot move after the button is pressed. The hardware emergency stop method allows the robot to move. : Locks/unlocks

the direction keys.

2.12 Charging Station Setup

After creating the map, place the robot on the charging station and add a marker with the charging station attribute. The marker name can be arbitrary, but the location attribute must be a charging station. The charging station must be within the map area and placed against a wall. Note that only one charging station should be added to each map. The area in front of the robot should be open within a 1.5m x 1.5m radius, free from interference from other charging station signals. In case

of positioning errors, manually pushing the robot to the charging station will automatically calibrate the robot's position based on the charging station's location.

Publishing the charging station location directly after adding it will automatically trigger the return-to-charging logic: the robot will navigate to the front of the charging station and then use infrared sensor information to perform backward

alignment. For versions below algorithm 4.5.0, when the robot is fully charged and the battery automatically protects against overcharge, the robot will perform a full-charge disengagement function, automatically moving forward 20cm. For algorithms 4.5.0 and above, the robot can perform overcharge protection while on the charging station and does not need to leave the station.

2.13 Emergency Stop Instructions

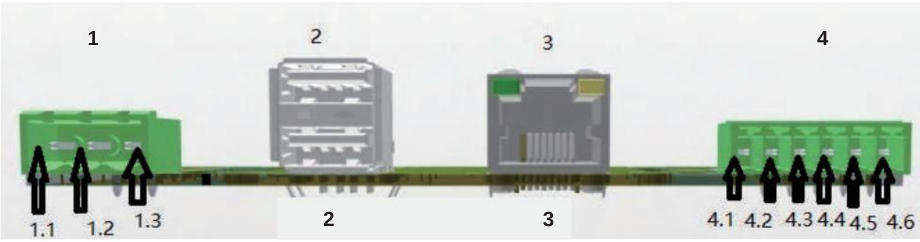
Hardware Emergency Stop: A hardware emergency stop switch on the robot itself. Pressing this switch allows the robot to move. Software Emergency

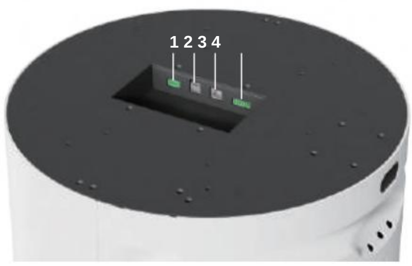
Stop: An algorithmic software switch; pressing this switch prevents the robot from moving.

Chassis External Expansion Interface Description

1. External power supply interface: 24V/10A, with overcurrent and short circuit protection, and current monitoring; 1.1 is positive, 1.3 is negative (as shown in the figure).

2: External USB expansion ports: 2 USB 2.0 expansion ports;





General chassis error code.

Chassis driver firmware version number: 4.XX

Note: You need to check the "About" - "Self-test Function" section of the self-developed deployment software.

Fault Codes	Explanation	Cause of the fault	Solution
1	Encoder ABZ alarm	a. Incorrect encoder wiring b. Damaged encoder c. Severe noise interference	a. Confirm the wiring is reliable and correct. b. Return to factory for repair. c. Keep away from high-current wiring.
2	Encoder UVW alarm	a. Incorrect encoder wiring b. Damaged encoder c. Severe noise interference	a. Confirm the wiring is reliable and correct. b. Return to factory for repair. c. Keep away from high-current wiring.
3	Poor location	a. Position command frequency too high b. Position loop gain too low c. Position tolerance setting too low d. Incorrect motor or encoder wiring e. Insufficient motor torque or excessive load	a. Confirm the wiring is reliable and correct. b. Return to factory for repair. c. Keep away from high-current wiring.
4	Stall	Motor speed too high	Decrease speed command
5	ADC zero-point anomaly	Motor current feedback channel abnormality	Return to factory for repair
6	Overload ABZ alarm	a. Overload b. Motor vibration c. Mechanical brake not released d. Incorrect wiring of motor and encoder	a. Replace with a higher-power driver and motor. b. Readjust the gain. c. Inspect the mechanical brake.
7	Power supply undervoltage ABZ	a. Power supply voltage too low b. Driver voltage measurement circuit malfunction	a. Check the power supply voltage. b. Return to factory for repair.
8	Power supply overvoltage	a. Power supply voltage too high b. Driver voltage measurement circuit malfunction c. Frequent motor start-stop under load	a. Check the power supply voltage. b. Return to factory for repair. c. Install a discharge module.

9	Overcurrent	a. The drive is damaged. b. Short circuit in motor power lines UVW c. Motor damage d. Automatic protection of power modules e. The servo motor vibrates violently.	a. Replace the driver b. Check the motor wiring c. Replace the motor d. Reconnect power e. Adjust servo gain parameters
10	Instantaneous discharge alarm	Excessive instantaneous discharge power	a. Check the mains voltage. b. Replace the driver c. Install a discharge module
11	Average Discharge Alarm	The average discharge power is too high.	a. Select a suitable regenerative discharge resistor b. Replace the driver c. Install a discharge module
12	Parameter read/write exception	Driver configuration parameter read/write error, return to factory for repair	
13	Input port redefinition	The input port function definition includes a function definition for repeatedly readjusting the input port.	
14	Line breakage protection	Communication failure	Check communication lines
15	Temperature alarm	Motor temperature too high	Check the lines and loads.

Chassis firmware version number: 3.XX Note:

This needs to be checked in the self-developed deployment software under ÿAboutÿ—ÿSelf-testing functionÿ.

Driver error status word *

Each bit represents a type of error, as follows: BIT[0]:

Internal error BIT[1]:

Encoder ABZ signal error BIT[2]: Encoder UVW

signal error BIT[3]: Encoder count error BIT[4]:

Driver temperature too high BIT[5]:

Driver bus voltage too high BIT[6]:

Driver bus voltage too low BIT[7]: Driver

output short circuit BIT[8]: Driver braking

resistor abnormal BIT[9]: Actual

tracking error exceeds the allowable value

BIT[10]: Reserved for

backup BIT[11]: I2*T fault (driver or motor overload)

BIT[12]: Reserved for

backup BIT[13]: Reserved

for backup BIT[14]: Find motor error (communication encoder)

BIT[15]: Reserved for future

use. Example: When the error code is 20480, convert the decimal 20480 to binary 0101 0000 0000 0000. Starting from the zeroth bit from right to left, the two 1s in the resulting error code are the 12th and 14th bits respectively. The error corresponding to 20480 is BIT[12] and BIT[14].

General standard output status

0: Command executed

successfully 10: Command

being processed 20: Mapping

in progress 30: Navigation in

progress 99: Unknown input command

Navigation related

600: Initialization pending

601: Running 602:

Navigation canceled

603: Navigation

successful 604:

Navigation failed 605:

Navigation rejected 606:

Navigation canceled 607:

Navigation resetting 608:

Navigation reset complete 609:

Navigation status lost

620: Configuration failed 621: Emergency stop button triggered

Version upgrade related

701: USB drive not found

702: Algorithm update package not found on USB drive.

703: Invalid update package in USB drive 704:

Update package corrupted 711:

No network 712: Failed

to download update package configuration file 713:

Failed to download update package

714: Failed to calculate update package checksum

715: Update package checksum failed 716:

Update server offline 717: Firmware

package search failed 762: Failed to

get network time 763: Failed to set system

time 788: Update package decompression

failed

Maintenance

The following maintenance plan outlines the process for regular cleaning and parts replacement.

1. Maintenance Tips

(1) Keep the equipment away from humid and high-temperature environments. (2) Avoid squeezing the equipment or hitting the equipment casing with sharp objects. (3) Avoid prolonged exposure to direct sunlight, as this will cause the plastic casing to age rapidly. (4) Keep the equipment's outer surface clean. When cleaning the equipment, disconnect all cables, close the chassis, and gently wipe the machine with a soft, slightly damp, lint-free cloth. Never use corrosive cleaning agents or chemicals to wipe the machine, especially the optical lens.

2. Clean the outer casing.

(1) Use only a lint-free soft cloth. Avoid using rough cloths, towels, paper towels, or similar items. (2) Avoid excessive rubbing, as this may cause damage. (3) Disconnect all external power sources, equipment, and cables during cleaning. (4) Keep the product away from liquids unless otherwise specified for the product. (5) Do not use sprays, bleach, or abrasives. (6) Do not spray cleaning agents directly onto the product.

3. Regularly inspected items

(1) Check if the equipment casing is loose or dirty.

(2) Check if there are any foreign objects wrapped around the chassis wheels or if there is excessive wear. (3) Check if the camera is clean and free of dirt. (4) Check if the lidar cover is clean and free of dirt. (5) Check if the ultrasonic radar mesh cover is dirty or obstructed.

Common problems and solutions

1. The casters are loose, causing abnormal movement.	Solution: Retighten and restore.
2. Incorrect charging station settings prevent the user from returning to the charging station.	Solution: Reconfigure the charging station
3. A loose laser connection cable prevents the new machine from scanning images.	Solution: Reconnect the laser connector cable tightly.
4. Suddenly unable to charge during use, or charging malfunction. 5.	Solution: Replace the charging station.
The machine runs erratically, spins in place, and does not follow the marked locations.	Solution: Recalibrate map points